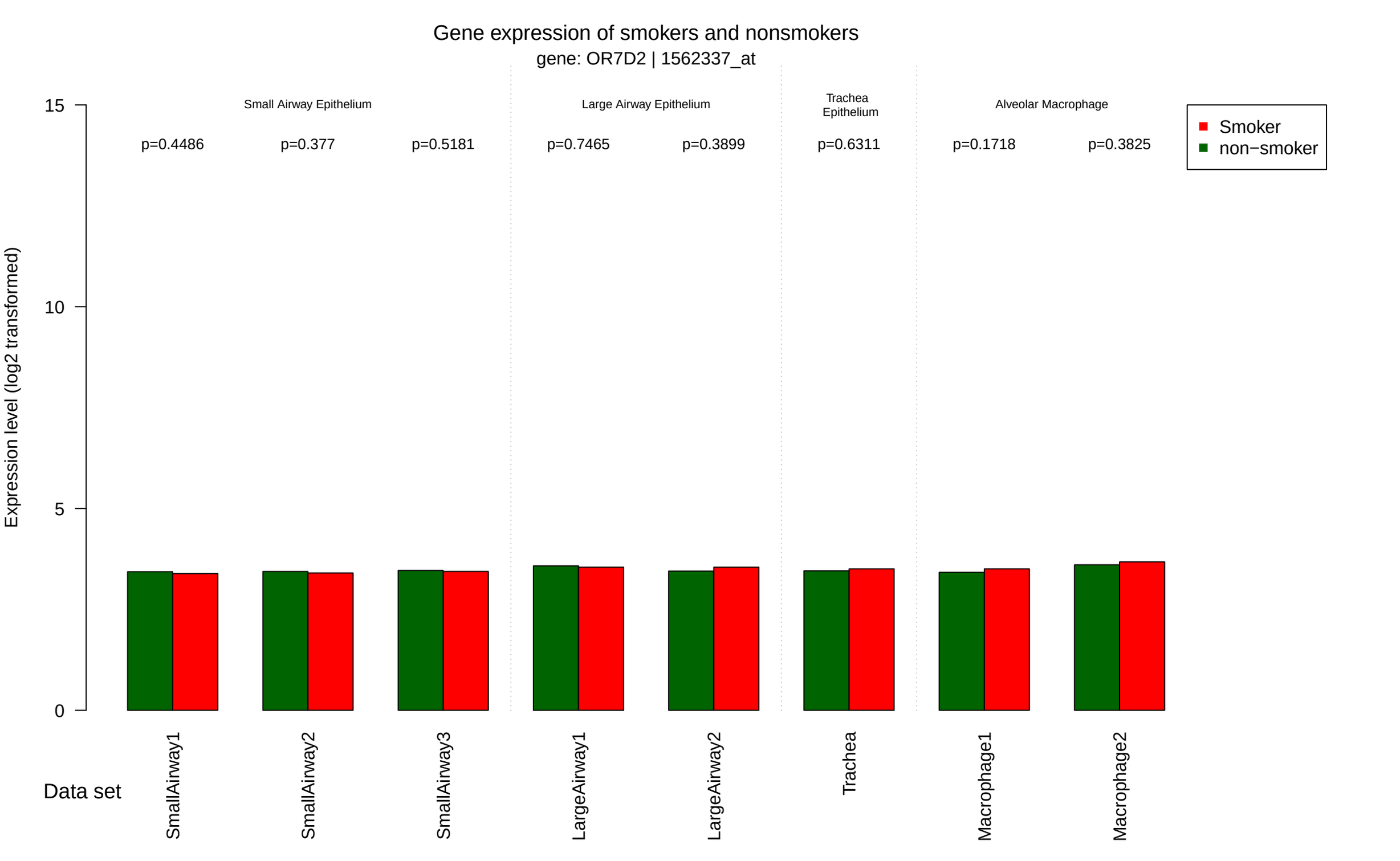
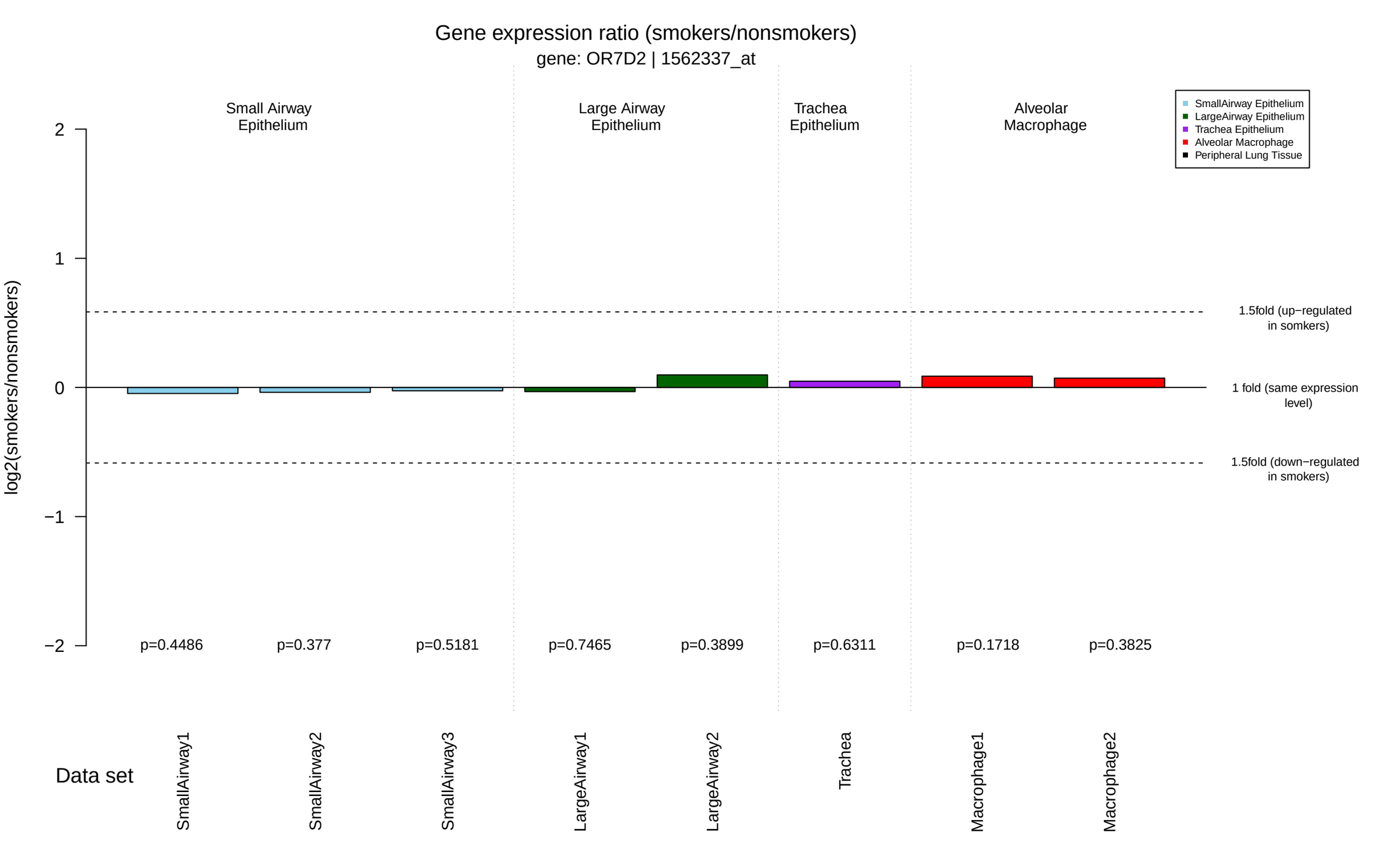
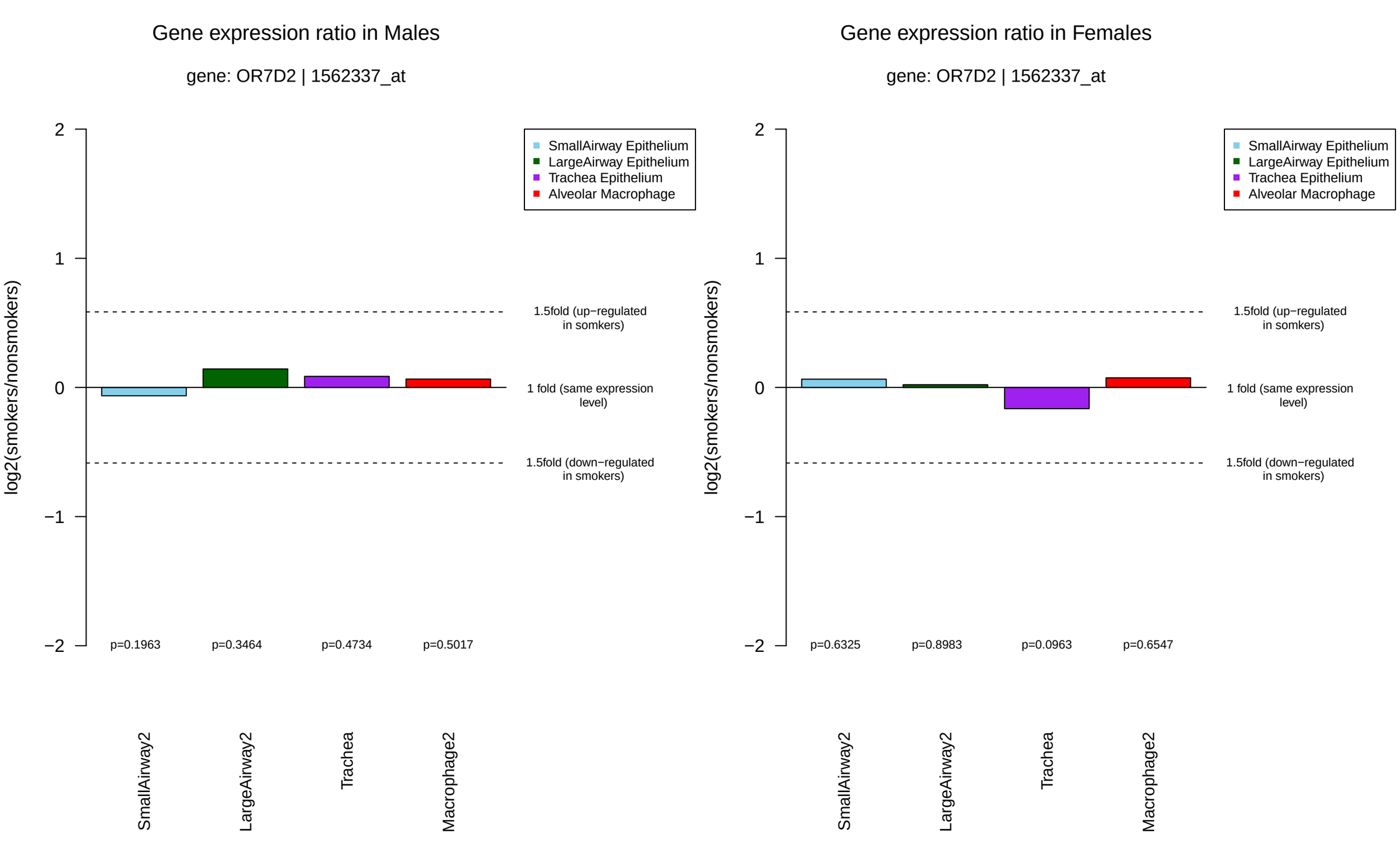
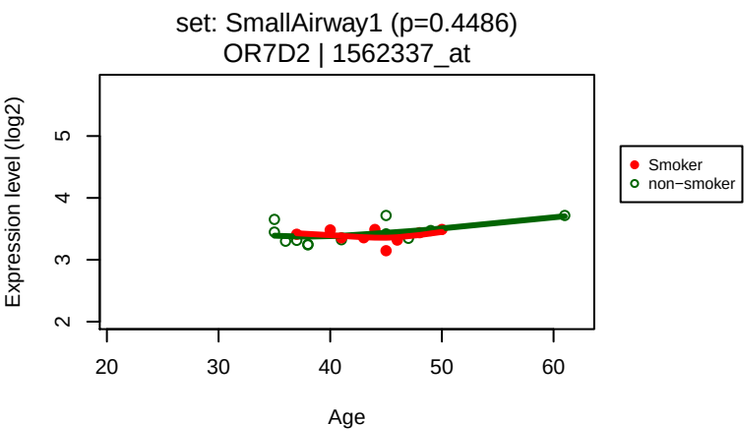
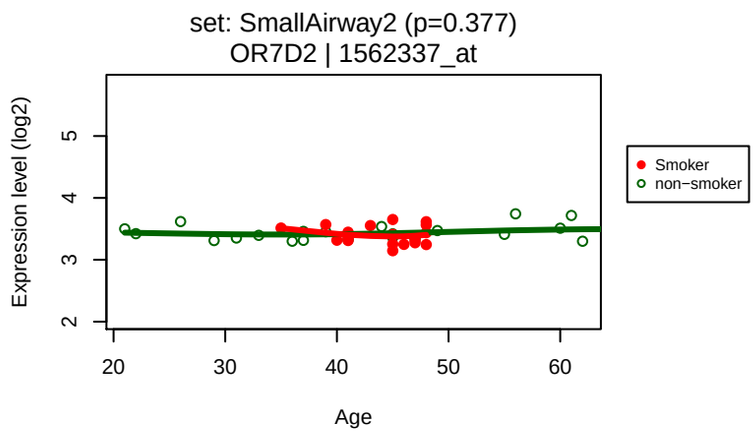
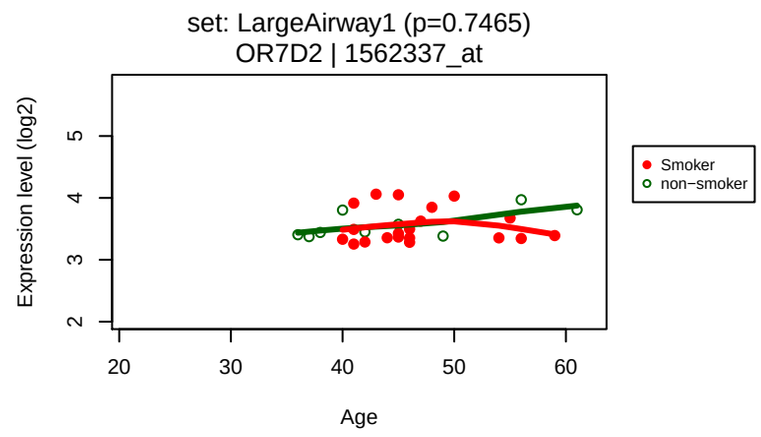

Gene expression of smokers and nonsmokers
gene: OR7D2 | 1562337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4486
p=0.377
p=0.5181
p=0.7465
p=0.3899
p=0.6311
p=0.1718
p=0.3825
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

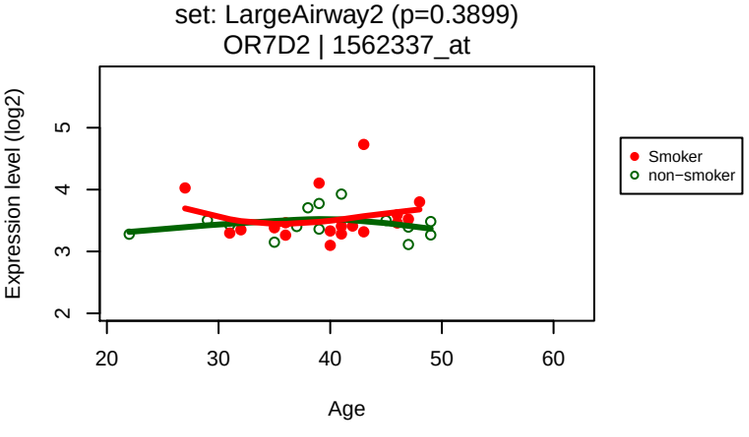

Gene expression ratio (smokers/nonsmokers)
gene: OR7D2 | 1562337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4486
p=0.377
p=0.5181
p=0.7465
p=0.3899
p=0.6311
p=0.1718
p=0.3825
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

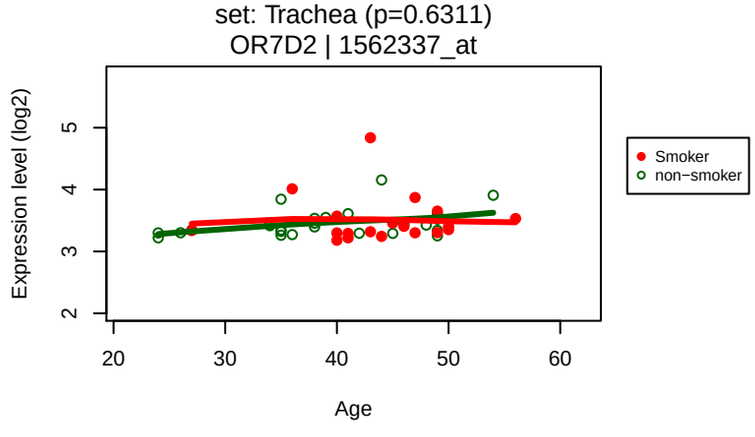

Gene expression ratio in Males
gene: OR7D2 | 1562337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1963
p=0.3464
p=0.4734
p=0.5017
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR7D2 | 1562337_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6325
p=0.8983
p=0.0963
p=0.6547
SmallAirway2
LargeAirway2
Trachea
Macrophage2

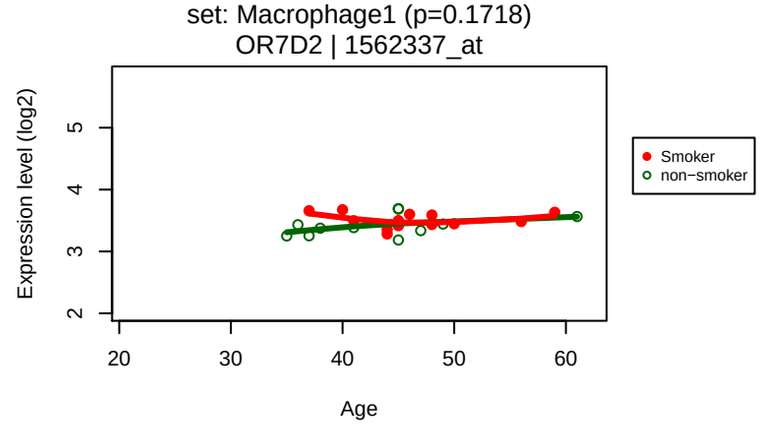

set: SmallAirway1 (p=0.4486)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker

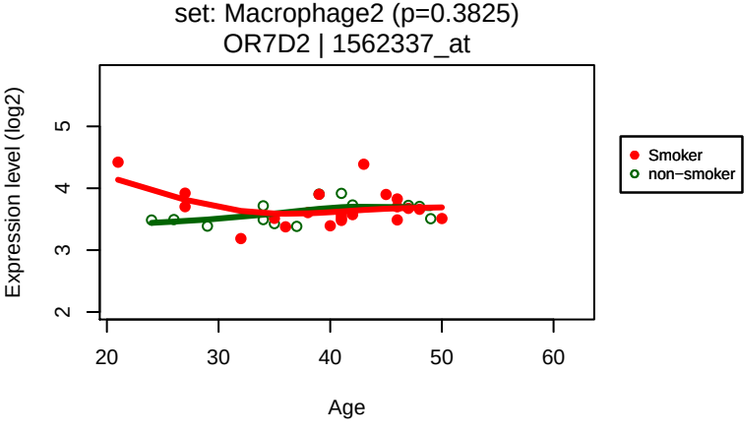

set: SmallAirway2 (p=0.377)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker

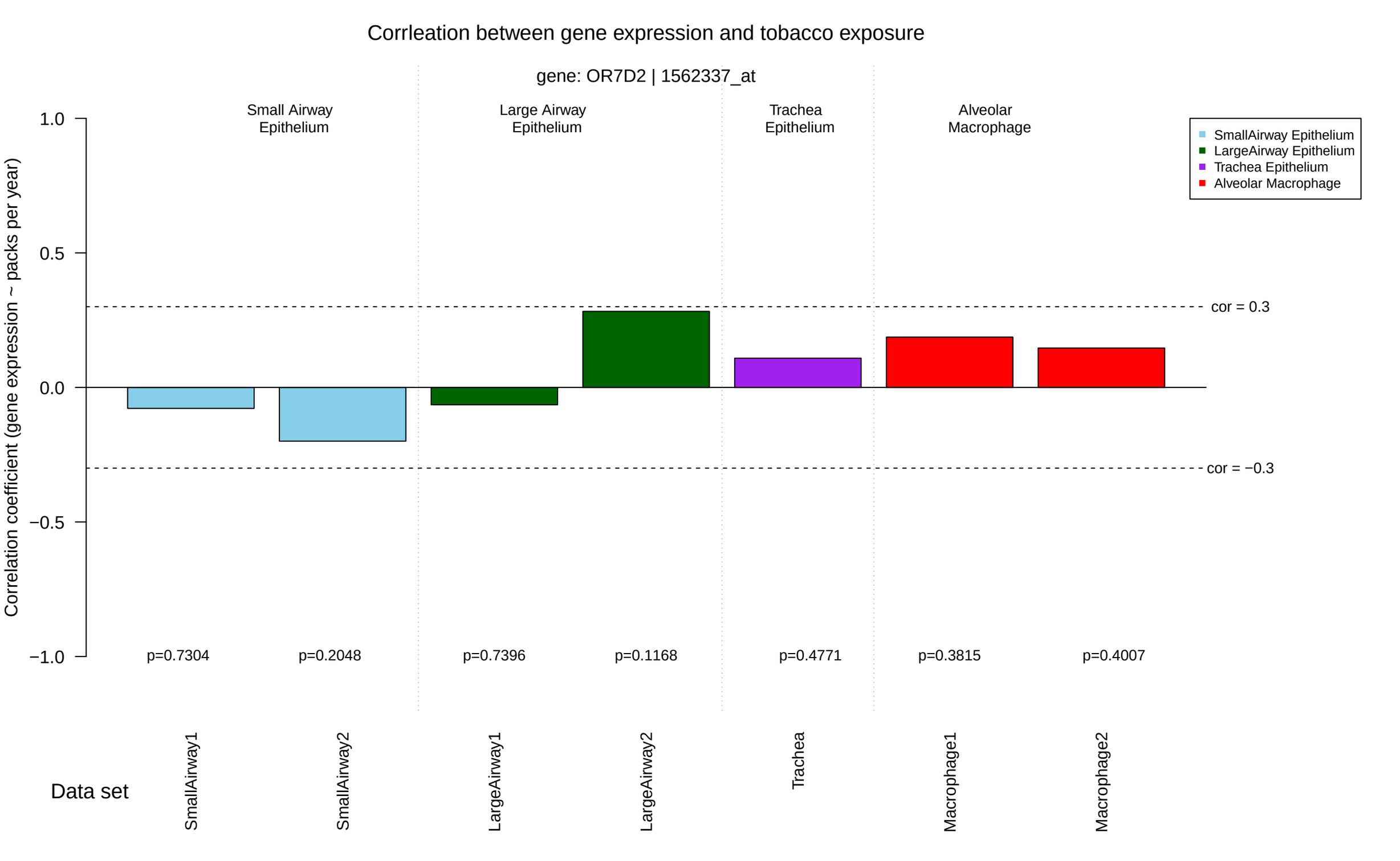

set: LargeAirway1 (p=0.7465)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3899)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6311)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1718)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3825)
OR7D2 | 1562337_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR7D2 | 1562337_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7304
p=0.2048
p=0.7396
p=0.1168
p=0.4771
p=0.3815
p=0.4007
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2